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1 DRONES FOR AGRICULTURE: Benefits And Concerns

➤ Pre-programmed automated application of materials not only leads to a perfect nutrient balance for crops, it also significantly saves wastage of these elements

Agricultural yield is directly proportional to the level of precision maintained in the materials used in farming, alongside quality of soil, climate and environmental conditions. India's agri-community is still dependent on their manual skills and ages of experience to maintain this level of precision. This often proves to be challenging, given the limitations and inconsistency in manual processes, which affect the yield many times.

Automation solution providers have taken this issue into account and have recommended robotic drones as the friend in need. But are our farmers ready for these?

The benefits

Agri-owners are well-aware of the problems that can crop up due to imbalances in various agricultural elements used. For instance, underuse of water leaves the crops dry, while its overuse can choke the crops, barring oxygen intake. Excessive pesticides can render the soil toxic. Underuse of fertilisers can lead to sub-optimal production rates. Drones are being designed to address these issues.

Pre-programmed automated application of materials leads to a perfect nutrient balance for crops. It also significantly saves wastage of these elements. Experts suggest that automation can save at least 20 per cent water wastage, and 15 per cent chemical wastage (weedicides, pesticides, etc).

Innovative drones are being designed to scan the health of soil from the sky. Crop-spraying drones use ultrasonic echoing or light detection and ranging (LIDAR) methods alongside field scans to position themselves at a proportionate distance from the field, and spray exactly the required amount of pesticides, weedicides and fertilisers.

On the other hand, irrigation drones use spectral or thermal sensors to recognise dry areas in farmlands

and spray the accurate volume of water required.

Drones can also generate geo-referenced 3D maps and other results like vegetation index, elevation maps, plantation counts, crop scouting, plant health status and so on. This enables farm owners to plan seeding and harvesting schedules, pay attention to plant health, understand soil conditions and reduce use of scarce or harmful chemicals.

Latest drones are being designed to even carry out plantations. These are being equipped with technology to shoot pods containing seeds and nutrients into the soil, after the farm has been scanned. While this technology is under R&D at the moment, it brings the potential to save up to 85 per cent of plantation costs through reduced manual labour, plantation equipment investments, land analysis costs and more.

The plans

India is banking heavily on smart farming technologies, and using drones is definitely on the list. Maharashtra government recently announced its plans to push drone-based agriculture. The state has approached IISc Bangalore and



Drones can be used for automated irrigation and chemicals distribution